A BURGERS–FISHER EQUATION FOR PRIME GAPS: SPECTRAL PROOF OF CONVERGENCE TO POISSON

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Abstract. We prove that the velocity field $u = -\partial_v \log f$, where f is the probability density of normalized prime gaps $v = g/\log x$, satisfies a Burgers–Fisher equation in a porous medium:

$$\partial_t u + 2v\, u\, \partial_v u = v\, \partial_v^2 u + (v{+}2)\, \partial_v u + u(1{-}u).$$

The Fisher–KPP reaction term $u(1-u)$ emerges naturally from the Fokker–Planck structure and drives u toward 1, which corresponds to the exponential (Poisson) distribution. The linearized perturbation operator has associated Laguerre polynomial eigenfunctions with spectral gap 1. Under a decaying sieve forcing, perturbations vanish as $O(1/(\log x)^{\beta})$. We verify numerically that the mutual information between consecutive gaps decays as $(\log x)^{-0.19}$—the first quantitative estimate of this rate—and that prime $p = 3$ carries 94% of the sieve signal.

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1. Introduction

Gallagher [1] proved, conditionally on the Hardy–Littlewood conjecture [2], that prime counts in short intervals converge to Poisson statistics. Despite decades of work [3, 4, 5], no quantitative convergence rate has been established.

We construct a velocity field from the prime gap distribution and show it satisfies a nonlinear PDE whose unique stable equilibrium is the Poisson regime. This provides both a mechanism (the spectral gap of a Laguerre-type operator) and, numerically, a rate ($\approx (\log x)^{-0.19}$).

2. The Burgers–Fisher Equation for Prime Gaps

2.1. The Feller process and its Fokker–Planck equation.

Consider the Feller diffusion (Cox–Ingersoll–Ross process)

$$dV = (1 - V)\,dt + \sqrt{2V}\,dW_t, \qquad V \geq 0, \tag{1}$$

with mean-reversion toward 1 and state-dependent noise $\sigma(V) = \sqrt{2V}$. Its Fokker–Planck equation for the density $f(v, t)$ is

$$\partial_t f = v f'' + (v + 1)f' + f, \tag{2}$$

where primes denote ∂_v.

Lemma 2.1. *$f_\infty(v) = e^{-v}$ is the unique probability density satisfying $\mathcal{L}f_\infty = 0$, where $\mathcal{L}f = vf'' + (v + 1)f' + f$.*

Proof. Direct verification: $\mathcal{L}(e^{-v}) = v \cdot e^{-v} + (v + 1)(-e^{-v}) + e^{-v} = 0$. Uniqueness follows from the general theory of Feller diffusions [7]: the process (1) with $2 \times$ mean-reversion speed$/\sigma^2 = 2 \cdot 1/2 = 1 \geq 1$ has an entrance boundary at 0 and a natural boundary at ∞, admitting a unique stationary distribution. □

2.2. Derivation of the velocity equation.

Theorem 2.2 (The sieve Burgers–Fisher equation)**.** *Let $f(v, t) > 0$ satisfy the Fokker–Planck equation (2) on $(0, \infty)$. Define the velocity field*

$$u(v, t) = -\frac{\partial}{\partial v} \log f(v, t) = -\frac{f'}{f}. \tag{3}$$

Then u satisfies the Burgers–Fisher equation in a porous medium:

$$\boxed{\partial_t u + 2v\,u\,\partial_v u = v\,\partial_v^2 u + (v{+}2)\,\partial_v u + u(1{-}u).} \tag{4}$$

Proof. Set $\psi = \log f$, so $\psi' = f'/f$ and $f''/f = \psi'' + (\psi')^2$. From (2):

$$\psi_t = v(\psi'' + \psi'^2) + (v + 1)\psi' + 1. \tag{5}$$

Differentiate (5) with respect to v:

$$\begin{aligned} \psi_t' &= \psi'' + v\psi''' + \psi'^2 + 2v\psi'\psi'' + \psi' + (v + 1)\psi'' \\ &= v\psi''' + (v + 2)\psi'' + 2v\psi'\psi'' + \psi'^2 + \psi'. \end{aligned}$$

Substitute $u = -\psi'$ (hence $\psi'' = -u'$, $\psi''' = -u''$):

$$-u_t = -vu'' - (v + 2)u' + 2vu\,u' + u^2 - u.$$

Multiply by -1 and rearrange:

$$u_t = vu'' + (v+2)u' - 2vu\,u' - u^2 + u = vu'' + (v+2)u' + u(1-u) - 2vu\,u'. \quad \square$$

Remark 2.3 (Structure of the equation)**.** Equation (4) combines three classical PDE mechanisms:

(1) *Burgers nonlinearity* $2vu\,u_v$: the hallmark of the Navier–Stokes and Burgers equations, arising here from the nonlinear transformation $u = -(\log f)'$.
(2) *Porous medium diffusion* $v\,u_{vv}$: the diffusivity equals v, so small gaps ($v \approx 0$) are "tight" and large gaps fluctuate freely.
(3) *Fisher–KPP reaction* $u(1-u)$: this term pushes u toward 1 from either side. It emerges from the Fokker–Planck structure, not imposed by hand.

3. Spectral Stability of the Poisson Equilibrium

Theorem 3.1 (Spectral analysis)**.**

[(i)]

(1) The unique bounded steady state of (4) is $u \equiv 1$, corresponding to the exponential distribution $f = e^{-v}$.
(2) The linearization around $u = 1$, writing $\delta u = u - 1$, is

$$\partial_t(\delta u) = v\,(\delta u)'' + (2-v)\,(\delta u)' - \delta u. \tag{6}$$

(3) The eigenvalue problem $v\,h'' + (2-v)\,h' + (\lambda+1)\,h = 0$ is the associated Laguerre equation with parameter $\alpha = 1$. Its solutions are the associated Laguerre polynomials $L_n^{(1)}(v)$ with eigenvalues

$$\lambda_n = -(n+1), \qquad n = 0, 1, 2, \ldots \tag{7}$$

(4) The spectral gap is $|\lambda_0| = 1$.

Proof. (i) A constant steady state $u = c$ satisfies $c(1-c) = 0$, giving $c = 0$ or $c = 1$. If $u = 0$ then $f' = 0$, i.e., $f = \text{const}$, which is not integrable on $[0, \infty)$. Thus $u = 1$ is the unique physically admissible steady state.

(ii) Substitute $u = 1 + \delta u$ into (4), keeping linear terms in δu:

$$\begin{aligned}(\delta u)_t + 2v \cdot 1 \cdot (\delta u)' &= v(\delta u)'' + (v+2)(\delta u)' + (1+\delta u)(1-1-\delta u)\\ &= v(\delta u)'' + (v+2)(\delta u)' - \delta u + O((\delta u)^2).\end{aligned}$$

Hence $(\delta u)_t = v(\delta u)'' + (2-v)(\delta u)' - \delta u$.

(iii) Setting $(\delta u)(v,t) = h(v)\,e^{\lambda t}$ gives $vh'' + (2-v)h' + (\lambda+1)h = 0$. With the substitution $\alpha = 1$ and $n = \lambda + 1$, this is the associated Laguerre equation $vh'' + (\alpha + 1 - v)h' + nh = 0$, whose polynomial solutions are $L_n^{(1)}(v)$ [6].

(iv) From $\lambda_n = -(n+1)$: $\lambda_0 = -1$, so all perturbation modes decay at rate ≥ 1. $\square$

4. Convergence Under Decaying Sieve Forcing

Theorem 4.1 (Perturbed convergence)**.** *Let $u(v,t)$ satisfy*

$$\partial_t u + 2vu\,u_v = vu_{vv} + (v+2)u_v + u(1-u) + \varepsilon(t)\,F(v), \tag{8}$$

where $\|F\|_\infty \le M$ *and* $|\varepsilon(t)| \le C_0/t^\beta$ *for* $\beta > 0$. *Then there exists* $C > 0$ *such that* $\|\delta u(\cdot,t)\|_{L^2(v\,e^{-v})} \le C/t^\beta$ *for all* t *sufficiently large.*

Proof. Define the energy $E(t) = \frac{1}{2}\|\delta u\|^2_{L^2(ve^{-v})}$, where $L^2(ve^{-v})$ is the natural weighted space for the associated Laguerre polynomials.

Step 1 (Dissipation). By the spectral gap (Theorem 3.1(iv)): $\langle \mathcal{L}_{\text{lin}}(\delta u), \delta u\rangle \le -|\lambda_0| \cdot \|\delta u\|^2 = -\|\delta u\|^2$.

Step 2 (Forcing). $|\langle \varepsilon F, \delta u\rangle| \le |\varepsilon| \cdot M' \cdot \|\delta u\|$, where $M' = \|F\|_{L^2(ve^{-v})}$. By Young's inequality: $|\varepsilon| M' \|\delta u\| \le \frac{1}{2}\|\delta u\|^2 + \frac{M'^2}{2}\varepsilon^2$.

Step 3 (Energy inequality). $\dot{E} \le -2E + E + \frac{M'^2}{2}\varepsilon^2 = -E + \frac{C_1}{t^{2\beta}}$.

Step 4 (Grönwall). The integrating factor e^t gives $E(t) \le E(t_0)e^{-(t-t_0)} + C_1 \int_{t_0}^t e^{-(t-\tau)}\tau^{-2\beta}\,d\tau$. The integral is $O(t^{-2\beta})$ by repeated integration by parts, yielding $E(t) = O(t^{-2\beta})$. □

5. Connection to Prime Gaps

Hypothesis 5.1 (Sieve diffusion)**.** Under the Hardy–Littlewood conjecture, the normalized gap density $f(v,s)$ at scale $s = \log x$ can be written as $f = e^{-v}[1 + \varphi(v,s)]$, where the perturbation φ is driven by the singular series and satisfies $\|\varphi(\cdot,s)\| = O(1/s^\beta)$ for some $\beta > 0$.

Under this hypothesis, the velocity field $u = 1 - \varphi'/(1+\varphi) \approx 1 - \varphi'$ satisfies a forced version of (4), and Theorem 4.1 gives $u \to 1$, i.e., convergence to Poisson.

Remark 5.2 (What the hypothesis does not give)**.** Hypothesis 5.1 concerns the marginal gap density. The mutual information $\text{MI}(g_n; g_{n+1})$—a joint quantity—requires additional input. We verify the MI decay numerically (§6) but leave the rigorous joint implication as an open problem.

5.1. The sieve as porous medium.

Under the Hardy–Littlewood conjecture: $f(v,s) \propto W(vs) \cdot e^{-v}$, where $W(g) = \prod_{p|g,\, p>2}(p-1)/(p-2)$. Each prime p is a "pore" with strength $\delta_p = 1/(p-2)$: $\delta_3 = 1$, $\delta_5 = 1/3$, $\delta_7 = 1/5$. As $s \to \infty$, gap values $g = vs$ grow and the mod-p structure equidistributes, weakening each pore.

5.2. Exact computation: the mod-3 coupling.

Proposition 5.3. *Under i.i.d. uniform prime residues mod* 3*:* $\text{MI}(g_n \bmod 3;\, g_{n+1} \bmod 3) = 1/4$ *bit, with structural zeros at* $(1,1)$ *and* $(2,2)$ *in the joint table.*

Proof. With $r_n = p_n \bmod 3 \in \{1,2\}$ i.i.d. uniform, the gap $g_n \equiv r_{n+1} - r_n \pmod 3$. Enumerating all 8 triples $(r_n, r_{n+1}, r_{n+2}) \in \{1,2\}^3$ yields the joint distribution:

$g_n \bmod 3 \setminus g_{n+1} \bmod 3$	0	1	2
0	2/8	1/8	1/8
1	1/8	**0**	1/8
2	1/8	1/8	**0**

Marginals: $P(0) = 1/2$, $P(1) = P(2) = 1/4$. Only entries $(1,2)$ and $(2,1)$ deviate from the product of marginals: $\text{MI} = 2 \times \frac{1}{8}\log_2(\frac{1/8}{1/4 \cdot 1/4}) = \frac{1}{4}$ bit. □

6. Numerical Verification

All measurements use $N = 200{,}000$ primes at 11 scales (10^3 to 10^8).

6.1. **MI decay rate.** $\mathrm{MI}_{\mathrm{total}}(x) \sim 0.555/(\log x)^{0.186}$, $R^2 = 0.986$. This is the first quantitative estimate of the convergence rate to Poisson.

6.2. **Prime 3 dominance.** $\mathrm{MI}_3 \,/\, \mathrm{MI}_{\mathrm{total}} \approx 94\%$ at all scales. Empirical $\mathrm{MI}_3 = 0.302$ bits vs. theoretical 0.250 bits; the 21% excess is the Lemke Oliver–Soundararajan bias [3], decaying as $O((\log\log x)/\log x)$ [5]. The structural zeros $P(1,1) = P(2,2) = 0$ hold exactly at all scales.

6.3. **Perturbation energy.** $E(s) = \|\varphi\|^2_{L^2(e^v)} \sim C/(\log x)^{0.63}$, $R^2 = 0.67$. The Laguerre decomposition shows the $n = 2$ mode (variance correction) dominates, while the $n = 1$ mode (mean shift) is nearly zero.

6.4. **Information cascade.**

Prime p	Decay exponent α_p	R^2	MI at 10^8
3	0.22	0.980	0.302 bits
7	3.83	0.996	0.006 bits
11	3.67	0.998	0.032 bits

Prime 3 ($\alpha \approx 0.22$) is the bottleneck; primes ≥ 7 decay as $(\log x)^{-3.5}$ or faster.

6.5. **Sieve diffusion model.** Starting from $f_0 = e^{-v}$ (Cramér) and iteratively adding sieve primes: adding $p = 3$ alone reduces KL divergence from the empirical distribution by 74% (from 0.061 to 0.016 nats at scale 10^7).

7. Open Problems

(1) **Derive** $\|\varphi(\cdot, s)\| = O(1/s^\beta)$ from the Hardy–Littlewood conjecture and equidistribution of primes in arithmetic progressions.
(2) **Prove** $\mathrm{MI} \to 0$ from the Burgers–Fisher framework (requires a joint version of Hypothesis 5.1).
(3) **Compute** the exponent $\alpha \approx 0.19$ from first principles: it should follow from the pore strength $\delta_3 = 1$ and the equidistribution rate of gaps mod 3.

Computational Details

Python 3 with SymPy, NumPy, SciPy. Runtime ≈ 2 minutes. Code, data, and interactive visualization: `https://github.com/tobiascanavesi/prime-fluid-dynamics`

Use of generative AI. Claude Opus 4.6 (Anthropic, 2025) assisted with code development and LaTeX formatting. All mathematical content is the original contribution of the author.

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